

Ronald Tetzlaff

professor

基本信息

姓名: Ronald Tetzlaff

所属院校: 德累斯顿工业大学

地区: Germany

年龄范围: 30岁以下

学科领域

生物医学工程

微电子科学与工程

电子与计算机工程

电子科学与技术

集成电路设计与集成系统

人工智能

研究领域

生物信号处理与模式识别

微电子器件与制造

纳米电子与先进工艺

电路与系统设计

医学影像处理与成像技术

深度学习与神经网络

信号处理与数字滤波

微纳电子与纳米技术

计算机体系结构

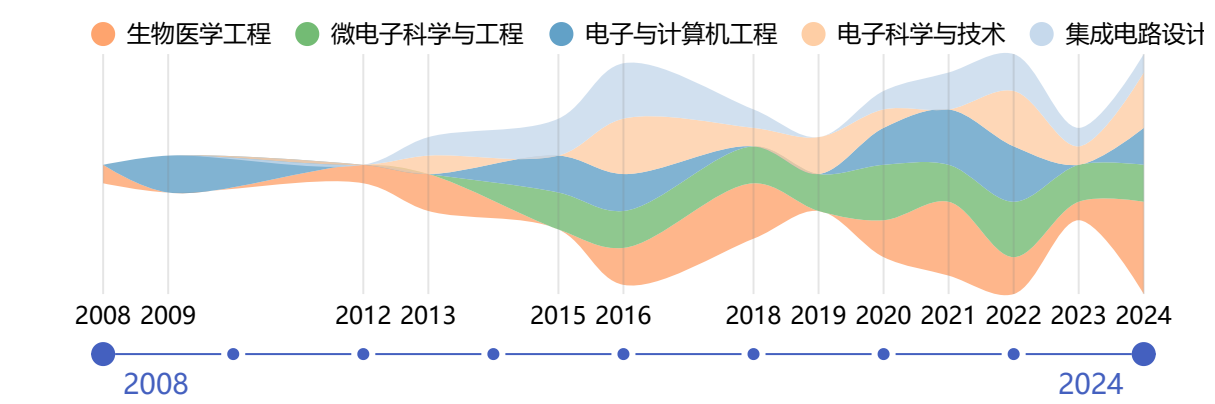
半导体器件物理与工艺

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个人简介

Prof. Dr. Ronald Tetzlaff is the Chief Officer of Technology Transfer and Internationalization at TU Dresden since August 2020. He also serves as a speaker for the Engineering Sciences at the university in 2020 and a member of the Steering Committee of the Else Kröner - Fresenius Center for Digital Health since 2019. He has held various academic positions at TU Dresden and other institutions. He was the dean of the Faculty of Electrical Engineering and Information Technology from 2016 - 2020. He has received several awards, including the Dresden Congress Award in 2014 and the Stahl - Innovationspreis in 2012. He is actively involved in editorial work and scientific committees.

近年发表的论文



Towards Generalizable Classification of Partial Discharges in Gas-Insulated HVDC Systems Using Neural Networks: Protrusions and Particles

IEEE TRANSACTIONS ON POWER DELIVERY

Seitz, Steffen; Goetz, Thomas; Lindenberg, Christopher; Tetzlaff, Ronald; Schlegel, Stephan

被引用次数: 1 2024

Analysis and Design of Bio-Inspired Circuits With Locally Active Memristors

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Ascoli, Alon; Demirkol, Ahmet Samil; Tetzlaff, Ronald; Chua, Leon

被引用次数: 8 2024

Deep Memristive Cellular Neural Networks for Image Classification and Segmentation

IEEE TRANSACTIONS ON NANOTECHNOLOGY

Horvath, Andras; Rajki, Franciska; Ascoli, Alon; Tetzlaff, Ronald

被引用次数: 2 2024

Real time hybrid medical image encryption algorithm combining memristor-based chaos with DNA coding

CHAOS SOLITONS & FRACTALS

Demirkol, Ahmet Samil; Sahin, Muhammet Emin; Karakaya, Baris; Ulutas, Hasan; Ascoli, Alon; Tetzlaff, Ronald

被引用次数: 3 2024

Modeling Excitable Cells with Memristors

JOURNAL OF LOW POWER ELECTRONICS AND APPLICATIONS

Sah, Maheshwar; Ascoli, Alon; Tetzlaff, Ronald; Rajamani, Vettriveeran; Budhathoki, Ram Kaji

被引用次数: 2 2024

High Frequency Response of Volatile Memristors

ADVANCED ELECTRONIC MATERIALS

Messarlis, Ioannis; Ascoli, Alon; Demirkol, Ahmet S.; Ntinis, Vasileios; Prousalis, Dimitrios; Tetzlaff, Ronald

被引用次数: 2 2024

Fault Impact Map for Memristive Crossbar Neural Networks

2024 13TH INTERNATIONAL CONFERENCE ON MODERN CIRCUITS AND SYSTEMS TECHNOLOGIES, MOCAST 2024

Schroedter, Richard; Haidar, Adnan; Tetzlaff, Ronald

被引用次数: 0

2024

Theoretical Analysis and Hardware Reproduction of Smale Paradox Based on CCM Neurons and Edge of Chaos

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Liang, Yan; Guo, Huimeng; Li, Yiqing; Jin, Peipei; Wang, Guangyi; Lu, Herbert Ho-Ching; Demirkol, Ahmet Samil; Tetzlaff, Ronald; Chen, Guanrong; Ascoli, Alon

被引用次数: 2

2024

An analytical approach to engineer multistability in the oscillatory response of a pulse-driven ReRAM

SCIENTIFIC REPORTS

Ascoli, Alon; Schmitt, Nicolas; Messaris, Ioannis; Demirkol, Ahmet Samil; Strachan, John Paul; Tetzlaff, Ronald; Chua, Leon

被引用次数: 5

2024

Local Activity Principle: Cause of Insulin Secretion by Pancreatic β -cells

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Sah, Maheshwar; Ascoli, Alon; Tetzlaff, Ronald; Rajamani, Vettriveeran; Budhathoki, Ram Kaji

被引用次数: 1

2024

Edge of Chaos Theory Sheds Light Into the All-to-None Phenomenon in Neurons-Part I: On the Fundamental Role of the Sodium Ion Channel

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Ascoli, Alon; Demirkol, Ahmet Samil; Tetzlaff, Ronald; Chua, Leon O.

被引用次数: 6

2024

Recent Advances and Future Prospects for Memristive Materials, Devices, and Systems

ACS NANO

Song, Min-Kyu; Kang, Ji-Hoon; Zhang, Xinyuan; Ji, Wonjae; Ascoli, Alon; Messaris, Ioannis; Demirkol, Ahmet Samil; Dong, Bowei; Aggarwal, Samarth; Wan, Weier; Hong, Seok-Man; Cardwell, Suma George; Boybat, Irem; Seo, Jae-sun; Lee, Jang-Sik; Lanza, Mario; Yeon, Hanwool; Onen, Murat; Li, Ju; Yildiz, Bilge; del Alamo, Jesus A.; Kim, Seyoung; Choi, Shinhyun; Milano, Gianluca; Ricciardi, Carlo; Alff, Lambert; Chai, Yang; Wang, Zhongrui; Bhaskaran, Harish; Hersam, Mark C.; Strukov, Dmitri; Wong, H. -S. Philip; Valov, Ilia; Gao, Bin; Wu, Huaqiang; Tetzlaff, Ronald; Sebastian, Abu; Lu, Wei; Chua, Leon; Yang, J. Joshua; Kim, Jeehwan

被引用次数: 52

2023

Exact Inductorless Realization of Chua Circuit Using Two Active Elements

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Demirkol, Ahmet S.; Ascoli, Alon; Weiher, Martin; Tetzlaff, Ronald

被引用次数: 7

2023

Self-Supervised Health Index Curve Generation for Condition-Based Predictive Maintenance

ELECTRONICS

Seitz, Steffen; Arnold, Marvin; Tetzlaff, Ronald; Holstein, Peter

被引用次数: 1

2023

Respiration and heartbeat motion correction of intraoperative thermographic images in neurosurgery

BIOMEDICAL SIGNAL PROCESSING AND CONTROL

Moshaei-Nezhad, Yahya; Tetzlaff, Ronald; Kirsch, Matthias

被引用次数: 0

2023

High Frequency Response of Non-Volatile Memristors

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Messaris, Ioannis; Ascoli, Alon; Demirkol, Ahmet S. S.; Tetzlaff, Ronald

被引用次数: 10

2023

A Compact Model of Threshold Switching Devices for Efficient Circuit Simulations

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Al Chawa, Mohamad Moner; Bedau, Daniel; Demirkol, Ahmet S.; Reiner, James W.; Stewart, Derek A.; Grobis, Michael K.; Tetzlaff, Ronald

被引用次数: 2

2023

Registration of IRT and visible light images in neurosurgery: analysis and comparison of automatic intensity-based registration approaches

INTERNATIONAL JOURNAL OF COMPUTER ASSISTED RADIOLOGY AND SURGERY

Moshaei-Nezhad, Yahya; Mueller, Juliane; Oelschlaegel, Martin; Kirsch, Matthias; Tetzlaff, Ronald

被引用次数: 1

2022

Graph Coloring via Locally-Active Memristor Oscillatory Networks

JOURNAL OF LOW POWER ELECTRONICS AND APPLICATIONS

Ascoli, Alon; Weiher, Martin; Herzig, Melanie; Slesazeck, Stefan; Mikolajick, Thomas; Tetzlaff, Ronald

被引用次数: 15

2022

Infrared Thermographic Imaging of Chest Wall Perfusion in Patients Undergoing Coronary Artery Bypass Grafting

ANNALS OF BIOMEDICAL ENGINEERING

Rasche, Stefan; Kleiner, Christian; Mueller, Jens; Rost, Antje; Ghazy, Tamer; Ploetze, Katrin; Tetzlaff, Ronald; Matschke, Klaus; Bota, Olimpiu

被引用次数: 1

2022

Physics-based modeling of a bi-layer Al₂O₃/Nb₂O₅ analog memristive device

2022 IEEE INTERNATIONAL SYMPOSIUM ON CIRCUITS AND SYSTEMS (ISCAS 22)

Schroedter, Richard; Mgeladze, Eter; Herzig, Melanie; Ascoli, Alon; Slesazeck, Stefan; Mikolajick, Thomas; Tetzlaff, Ronald

被引用次数: 1

2022

Performance Analysis of Memristive-CNN based on a VCM Device Model

2022 IEEE INTERNATIONAL SYMPOSIUM ON CIRCUITS AND SYSTEMS (ISCAS 22)

Wang, Yongmin; Ascoli, Alon; Tetzlaff, Ronald; Rana, Vikas; Menzel, Stephan

被引用次数: 0

2022

Guest Editorial Memristive Circuits and Systems for Edge-Computing Applications

IEEE JOURNAL ON EMERGING AND SELECTED TOPICS IN CIRCUITS AND SYSTEMS

Sirakoulis, Georgios Ch.; Ascoli, Alon; Tetzlaff, Ronald; Yu, Shimeng

被引用次数: 0

2022

Edge of Chaos Explains Prigogines Instability of the Homogeneous

IEEE JOURNAL ON EMERGING AND SELECTED TOPICS IN CIRCUITS AND SYSTEMS

Ascoli, Alon; Demirkol, Ahmet S.; Tetzlaff, Ronald; Chua, Leon O.

被引用次数: 9

2022

Pulse and Breathing Motion Artifacts Correction of Intraoperative Thermal Imaging in Neurosurgery

2022 IEEE INTERNATIONAL SYMPOSIUM ON CIRCUITS AND SYSTEMS (ISCAS 22)

Moshaei-Nezhad, Yahya; Oelschlagel, Martin; Mueller, Juliane; Kirsch, Matthias; Tetzlaff, Ronald

被引用次数: 0

2022

An Analog Memristive and Memcapacitive Device for Neuromorphic Computing

2022 29TH IEEE INTERNATIONAL CONFERENCE ON ELECTRONICS, CIRCUITS AND SYSTEMS (IEEE ICECS 2022)

Mgeladze, Eter; Herzig, Melanie; Schroedter, Richard; Tetzlaff, Ronald; Mikolajick, Thomas; Slesazeck, Stefan

被引用次数: 1

2022

Toward Simplified Physics-Based Memristor Modeling of Valence Change Mechanism Devices

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Ntinis, Vasileios; Ascoli, Alon; Messaris, Ioannis; Wang, Yongmin; Rana, Vikas; Menzel, Stephan; Tetzlaff, Ronald

被引用次数: 14

2022

A Compact and Continuous Reformulation of the Strachan TaOx Memristor Model With Improved Numerical Stability

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Demirkol, Ahmet S.; Ascoli, Alon; Messaris, Ioannis; Al Chawa, Mohamad Moner; Tetzlaff, Ronald; Chua, Leon O.

被引用次数: 6

2022

Edge of Chaos Theory Resolves Smale Paradox

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Ascoli, Alon; Demirkol, Ahmet Samil; Tetzlaff, Ronald; Chua, Leon

被引用次数: 41

2022

Pattern formation dynamics in a Memristor Cellular Nonlinear Network structure with a numerically stable VO2 memristor model

JAPANESE JOURNAL OF APPLIED PHYSICS

Demirkol, Ahmet Samil; Ascoli, Alon; Messaris, Ioannis; Tetzlaff, Ronald

被引用次数: 10

2022

Edge of Chaos Is Sine Qua Non for Turing Instability

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Ascoli, Alon; Demirkol, Ahmet Samil; Tetzlaff, Ronald; Chua, Leon

被引用次数: 21

2022

A Deep Study of Resistance Switching Phenomena in TaOx ReRAM Cells: System-Theoretic Dynamic Route Map Analysis and Experimental Verification

ADVANCED ELECTRONIC MATERIALS

Ascoli, Alon; Menzel, Stephan; Rana, Vikas; Kempen, Tim; Messaris, Ioannis; Demirkol, Ahmet Samil; Schulten, Michael; Siemon, Anne; Tetzlaff, Ronald

被引用次数: 17

2022

Registration and Fusion of Visible Light and IRT Images in Neurosurgery

2021 IEEE BIOMEDICAL CIRCUITS AND SYSTEMS CONFERENCE (IEEE BIOCAS 2021)

Moshaei-Nezhad, Yahya; Mueller, Juliane; Oelschlaegel, Martin; Kirsch, Matthias; Tetzlaff, Ronald

被引用次数: 1

2021

Control Strategies to Optimize Graph Coloring via M-CNNs with Locally-Active NbOx Memristors

2021 10TH INTERNATIONAL CONFERENCE ON MODERN CIRCUITS AND SYSTEMS TECHNOLOGIES (MOCAST)

Ascoli, Alon; Weiher, Martin; Tetzlaff, Ronald; Herzig, Melanie; Slesazeck, Stefan; Mikolajick, Thomas

被引用次数: 2

2021

NbO2-Mott Memristor: A Circuit-Theoretic Investigation

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Messaris, Ioannis; Brown, Timothy D.; Demirkol, Ahmet S.; Ascoli, Alon; Al Chawa, M. Moner; Williams, R. Stanley; Tetzlaff, Ronald; Chua, Leon O.

被引用次数: 30

2021

How to Build a Memristive Integrate-and-Fire Model for Spiking Neuronal Signal Generation

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Kang, Sung Mo; Choi, Donguk; Eshraghian, Jason K.; Zhou, Peng; Kim, Jieun; Kong, Bai-Sun; Zhu, Xiaojian; Demirkol, Ahmet Samil; Ascoli, Alon; Tetzlaff, Ronald; Lu, Wei D.; Chua, Leon O.

被引用次数: 27

2021

Motion correction for IRT imaging in neurosurgery: Analysis and comparison of frequency-/filter- and intensity-based approaches

INFRARED PHYSICS & TECHNOLOGY

Moshaei-Nezhad, Yahya; Mueller, Juliane; Schnabel, Christian; Kirsch, Matthias; Tetzlaff, Ronald

被引用次数: 4

2021

On Local Activity and Edge of Chaos in a NaMLab Memristor

FRONTIERS IN NEUROSCIENCE

Ascoli, Alon; Demirkol, Ahmet S.; Tetzlaff, Ronald; Slesazeck, Stefan; Mikolajick, Thomas; Chua, Leon O.

被引用次数: 59

2021

A robust optical flow motion estimation and correction method for IRT imaging in brain surgery

QUANTITATIVE INFRARED THERMOGRAPHY JOURNAL

Moshaei-Nezhad, Yahya; Mueller, Juliane; Schnabel, Christian; Kirsch, Matthias; Tetzlaff, Ronald

被引用次数: 8

2021

A Compact Memristor Model for Neuromorphic ReRAM Devices in Flux-Charge Space

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Al Chawa, Mohamad Moner; Picos, Rodrigo; Tetzlaff, Ronald

被引用次数: 7

2021

Improved Vertex Coloring With NbOx Memristor-Based Oscillatory Networks

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Weiher, Martin; Herzig, Melanie; Tetzlaff, Ronald; Ascoli, Alon; Mikolajick, Thomas; Slesazeck, Stefan

被引用次数: 17

2021

Motion Correction in Multimodal Intraoperative Imaging

IEEE TRANSACTIONS ON BIOMEDICAL CIRCUITS AND SYSTEMS

Chen, Fang; Mueller, Jan; Mueller, Jens; Mueller, Juliane; Kirsch, Matthias; Tetzlaff, Ronald

被引用次数: 4

2020

Experimental evaluation of the dynamic route map in the reset transition of memristive ReRAMs

CHAOS SOLITONS & FRACTALS

Maldonado, David; Gonzalez, Mireia B.; Campabadal, Francesca; Jimenez-Molinos, Francisco; Al Chawa, M. Moner; Stavrinides, Stavros G.; Roldan, Juan B.; Tetzlaff, Ronald; Picos, Rodrigo; Chua, Leon O.

被引用次数: 19

2020

Intrinsic plasticity of silicon nanowire neurotransistors for dynamic memory and learning functions

NATURE ELECTRONICS

Baek, Eunhye; Das, Nikhil Ranjan; Cannistraci, Carlo Vittorio; Rim, Taiuk; Bermudez, Gilbert Santiago Canon; Nych, Khrystyna; Cho, Hyeonsu; Kim, Kihyun; Baek, Chang-Ki; Makarov, Denys; Tetzlaff, Ronald; Chua, Leon; Baraban, Larysa; Cuniberti, Gianaurelio

被引用次数: 35

2020

Theoretical Foundations of Memristor Cellular Nonlinear Networks: A DRM2-Based Method to Design Memcomputers With Dynamic Memristors

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Ascoli, Alon; Tetzlaff, Ronald; Kang, Sung-Mo; Chua, Leon O.

被引用次数: 46

2020

Theoretical Foundations of Memristor Cellular Nonlinear Networks: Stability Analysis With Dynamic Memristors

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Ascoli, Alon; Messaris, Ioannis; Tetzlaff, Ronald; Chua, Leon O.

被引用次数: 41

2020

Theoretical Foundations of Memristor Cellular Nonlinear Networks: Memcomputing With Bistable-Like Memristors

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Tetzlaff, Ronald; Ascoli, Alon; Messaris, Ioannis; Chua, Leon O.

被引用次数: 52

2020

Multiple slopes in the negative differential resistance region of NbOx-based threshold switches

JOURNAL OF PHYSICS D-APPLIED PHYSICS

Herzig, Melanie; Weiher, Martin; Ascoli, Alon; Tetzlaff, Ronald; Mikolajick, Thomas; Slesazeck, Stefan

被引用次数: 22

2019

Pattern Formation With Locally Active S-Type NbOx Memristors

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Weiher, Martin; Herzig, Melanie; Tetzlaff, Ronald; Ascoli, Alon; Mikolajick, Thomas; Slesazeck, Stefan

被引用次数: 49

2019

Improvement of NbOx-based threshold switching devices by implementing multilayer stacks

SEMICONDUCTOR SCIENCE AND TECHNOLOGY

Herzig, Melanie; Weiher, Martin; Ascoli, Alon; Tetzlaff, Ronald; Mikolajick, Thomas; Slesazeck, Stefan

被引用次数: 12

2019

An Improved Cellular Nonlinear Network Architecture for Binary and Grayscale Image Processing

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Mueller, Jens; Wittig, Robert; Mueller, Jan; Tetzlaff, Ronald

被引用次数: 6

2018

Intraoperative motion correction in neurosurgery: a comparison of intensity- and feature-based methods

BIOMEDICAL ENGINEERING-BIOMEDIZINISCHE TECHNIK

Chen, Fang; Mueller, Jan; Mueller, Jens; Mueller, Juliane; Boehl, Elisa; Kirsch, Matthias; Tetzlaff, Ronald

被引用次数: 1

2018

Guest Editorial - Special Issue on 'Memristors: Devices, Models, Circuits, Systems, and Applications'

INTERNATIONAL JOURNAL OF CIRCUIT THEORY AND APPLICATIONS

Tetzlaff, Ronald; Corinto, Fernando; Picos, Rogrigo; Ogorzalek, Maciej

被引用次数: 0

2018

Ultrasensitive detection of Ebola matrix protein in a memristor mode

NANO RESEARCH

Ibarlucea, Bergoi; Akbar, Teuku Fawzul; Kim, Kihyun; Rim, Taiuk; Baek, Chang-Ki; Ascoli, Alon; Tetzlaff, Ronald; Baraban, Larysa; Cuniberti, Gianaurelio

被引用次数: 39

2018

Special Issue on Memristors, Cellular Neural Networks, and Chua's Circuit, in Honor of Leon O. Chua's 80th Birthday

IEEE CIRCUITS AND SYSTEMS MAGAZINE

Tetzlaff, Ronald; Biolek, Dalibor

被引用次数: 0

2018

Registration and Fusion of Thermographic and Visual-Light Images in Neurosurgery

IEEE TRANSACTIONS ON BIOMEDICAL CIRCUITS AND SYSTEMS

Mueller, Jan; Mueller, Jens; Chen, Fang; Tetzlaff, Ronald; Mueller, Juliane; Boehl, Elisa; Kirsch, Matthias; Schnabel, Christian

被引用次数: 13

2018

Analysis of memristors with nonlinear memristance versus state maps

INTERNATIONAL JOURNAL OF CIRCUIT THEORY AND APPLICATIONS

Biolek, Zdenek; Biolek, Dalibor; Biolkova, Viera; Kolka, Zdenek; Ascoli, Alon; Tetzlaff, Ronald

被引用次数: 8

2017

New Signal Processing Methods for the Development of Seizure Warning Devices in Epilepsy

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Senger, Vanessa; Tetzlaff, Ronald

被引用次数: 10

2016

NEROvideo: a general-purpose CNN-UM video processing system

JOURNAL OF REAL-TIME IMAGE PROCESSING

Mueller, Jens; Mueller, Jan; Tetzlaff, Ronald

被引用次数: 5

2016

History Erase Effect in a Non-Volatile Memristor

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Ascoli, Alon; Tetzlaff, Ronald; Chua, Leon O.; Strachan, John Paul; Williams, Richard Stanley

被引用次数: 59

2016

The First Ever Real Bistable Memristors-Part I: Theoretical Insights on Local Fading Memory

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Ascoli, Alon; Tetzlaff, Ronald; Chua, Leon O.

被引用次数: 27

2016

The First Ever Real Bistable Memristors-Part II: Design and Analysis of a Local Fading Memory System

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Ascoli, Alon; Tetzlaff, Ronald; Chua, Leon O.

被引用次数: 28

2016

A class of versatile circuits, made up of standard electrical components, are memristors

INTERNATIONAL JOURNAL OF CIRCUIT THEORY AND APPLICATIONS

Ascoli, Alon; Corinto, Fernando; Tetzlaff, Ronald

被引用次数: 39

2016

Generalized boundary condition memristor model

INTERNATIONAL JOURNAL OF CIRCUIT THEORY AND APPLICATIONS

Ascoli, Alon; Corinto, Fernando; Tetzlaff, Ronald

被引用次数: 50

2016

A Cellular Network Architecture With Polynomial Weight Functions

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Mueller, Jens; Mueller, Jan; Braunschweig, Robert; Tetzlaff, Ronald

被引用次数: 4

2016

Robust Simulation of a TaO Memristor Model

RADIOENGINEERING

Ascoli, Alon; Tetzlaff, Ronald; Chua, Leon

被引用次数: 20

2015

Nonlinear Dynamics of a Locally-Active Memristor

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Ascoli, Alon; Slesazeck, Stefan; Maehne, Hannes; Tetzlaff, Ronald; Mikolajick, Thomas

被引用次数: 153

2015

The Art of Finding Accurate Memristor Model Solutions

IEEE JOURNAL ON EMERGING AND SELECTED TOPICS IN CIRCUITS AND SYSTEMS

Ascoli, Alon; Tetzlaff, Ronald; Biolek, Zdenek; Kolka, Zdenek; Biolkova, Viera; Biolek, Dalibor

被引用次数: 53

2015

Synchronization conditions in simple memristor neural networks

JOURNAL OF THE FRANKLIN INSTITUTE-ENGINEERING AND APPLIED MATHEMATICS

Ascoli, Alon; Lanza, Valentina; Corinto, Fernando; Tetzlaff, Ronald

被引用次数: 28

2015

Neuronal Synapse as a Memristor: Modeling Pair- and Triplet-Based STDP Rule

IEEE TRANSACTIONS ON BIOMEDICAL CIRCUITS AND SYSTEMS

Cai, Weiran; Ellinger, Frank; Tetzlaff, Ronald

被引用次数: 42

2015

**Special Issue Memristor Theory, Modeling and Simulation within the COST action IC1401
Memristors - Devices, Models, Circuits, Systems and Applications (MemoCiS)**

RADIOENGINEERING

Biolek, Dalibor; Tetzlaff, Ronald

被引用次数: 0

2015

Adaptive Neuromorphic Architecture (ANA)

NEURAL NETWORKS

Wang, Frank Zhigang; Chua, Leon O.; Yang, Xiao; Na Helian; Tetzlaff, Ronald; Schmidt, Torsten; Li, Caroline; Garcia Carrasco, Jose Manuel; Chen, Wanlong; Chu, Dominique

被引用次数: 42

2013

Memristor Model Comparison

IEEE CIRCUITS AND SYSTEMS MAGAZINE

Ascoli, Alon; Corinto, Fernando; Senger, Vanessa; Tetzlaff, Ronald

被引用次数: 145

2013

Untitled

IEEE CIRCUITS AND SYSTEMS MAGAZINE

Tetzlaff, Ronald; Wu, Chai Wah

被引用次数: 0

2013

The Seizure Prediction Problem in Epilepsy: Cellular Nonlinear Networks (vol 12, pg 8, 2012)

IEEE CIRCUITS AND SYSTEMS MAGAZINE

Tetzlaff, Ronald; Senger, Vanessa

被引用次数: 1

2013

The Seizure Prediction Problem in Epilepsy: Cellular Nonlinear Networks

IEEE CIRCUITS AND SYSTEMS MAGAZINE

Tetzlaff, Ronald; Senger, Vanessa

被引用次数: 13

2012

Special issue on cellular wave computing architectures, Part II

INTERNATIONAL JOURNAL OF CIRCUIT THEORY AND APPLICATIONS

Roska, Tamas; Arena, Paolo; Lin, Chin-Teng; Tetzlaff, Ronald

被引用次数: 2

2009

CNN Technology for Spatiotemporal Signal Processing

EURASIP JOURNAL ON ADVANCES IN SIGNAL PROCESSING

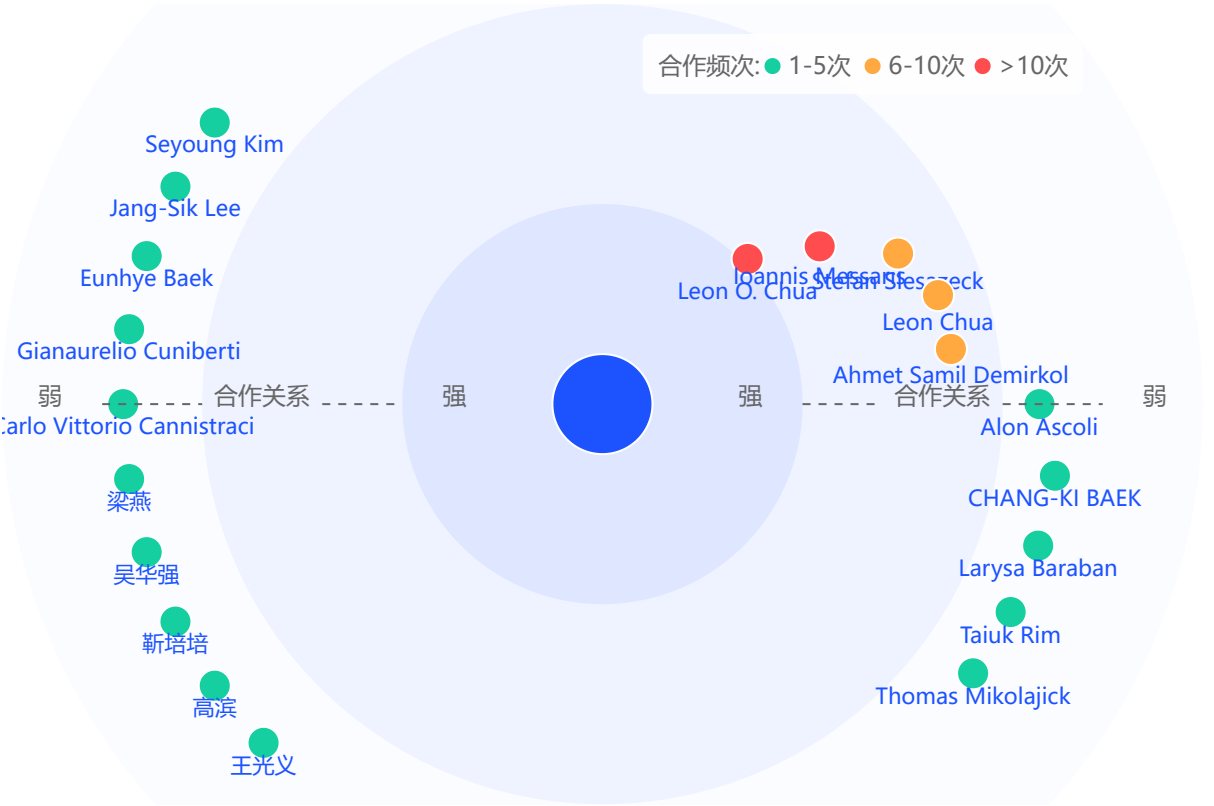
Lopez Vilarino, David; Cabello Ferrer, Diego; Brea Sanchez, Victor Manuel; Tetzlaff, Ronald; Lin, Chin-Teng

被引用次数: 0

2009

Toward an autonomous platform for spatio-temporal EEG-signal analysis based on cellular nonlinear networks

学术合作



Leon O. Chua	合作次数: 14
加州大学-伯克利	
Ioannis Messaris	合作次数: 11
德累斯顿工业大学	
Stefan Slesazec	合作次数: 8
德累斯顿工业大学	
Leon Chua	合作次数: 7
加州大学-伯克利	
Ahmet Samil Demirkol	合作次数: 7
德累斯顿工业大学	

还有 37 个合作者